Promethazine

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Continuing Education Activity

Promethazine is a medication used to manage and treat allergic conditions, nausea and vomiting, motion sickness, and sedation. It also has off-label uses for nausea and vomiting in pregnancy. Promethazine is a phenothiazine derivative with antidopaminergic, antihistamine, and anticholinergic properties. This activity reviews the FDA approved-indications, mechanism of action, administration, dosing, adverse drug reactions, contraindications, warning, precautions, monitoring, and toxicity of promethazine. In addition, this activity will highlight the pertinent interprofessional roles of the interprofessional team during the use of promethazine n the treatment of patients with allergic conditions, nausea and vomiting, motion sickness, and related conditions.

Objectives:

- Identify the mechanism of action of promethazine.
- Describe the adverse effects of promethazine.
- Review the appropriate monitoring for toxicity of promethazine.
- Summarize interprofessional team strategies for improving care coordination and communication to advance promethazine in patients who present with conditions where it has therapeutic value and can improve outcomes.

Access free multiple choice questions on this topic.

Indications

FDA-approved Indications

- Allergic conditions: Promethazine is a first-generation antihistamine, and thus it is indicated for a variety of allergic conditions, including seasonal allergic rhinitis and allergic conjunctivitis, and uncomplicated skin manifestations of urticaria, angioedema.[1][2] It is also indicated as adjunctive therapy to epinephrine for anaphylactic reactions.[3]
- Nausea and vomiting: Phenothiazines such as promethazine have substantial antiemetic activity. Clinicians use promethazine to control nausea and vomiting associated with anesthesia or chemotherapy.[4][5] It is commonly used postoperatively as an antiemetic. The antiemetic activity increases with increased dosing; however, side effects also increase, which often limits maximal dosing.[6]
- Motion sickness: Promethazine can serve as prophylactic therapy for motion sickness. It is most effective when given 30 minutes to 1 hour before the triggering event.[7]
- Sedation: Promethazine can be used as adjunctive therapy with other analgesics to provide preoperative, postoperative, or obstetric sedation.[8]

Off-label Usage

- Nausea and vomiting associated with pregnancy: According to the American College of Obstetrics, promethazine is useful in nausea and vomiting during pregnancy when preferred agents do not provide symptomatic relief.[9] A double-blinded randomized controlled trial of intravenous promethazine versus metoclopramide in women with hyperemesis gravidarum found similar efficacy between the two drugs in controlling nausea and vomiting at 24 hours but found increased side effects in the promethazine group such as dry mouth, drowsiness, dizziness, and dystonia. Research has also identified phenothiazines as a potential cause of congenital malformations in one study, but other studies attest to their safety. Promethazine is classified as category C by the FDA. Alternative therapies may be indicated before promethazine, considering the risks to the fetus and adverse effects on the mother.[10]

Mechanism of Action

Promethazine is a phenothiazine derivative with antidopaminergic, antihistamine, and anticholinergic properties. Other phenothiazine derivatives include prochlorperazine and chlorpromazine. Promethazine works as a direct antagonist at the mesolimbic dopamine receptors and alpha-adrenergic receptors in the brain. Promethazine exhibits its antihistamine effects as an H1-receptor blocker.[11][12]

Administration

Promethazine has several routes of administration, including oral, rectal, intramuscular, and intravenous. The tablets, solutions, and suppository dosage is generally 12.5 mg to 50 mg.

- Promethazine hydrochloride injection solution: 25 mg/mL; 50 mg/mL
- Promethazine hydrochloride oral solsution: 6.25 mg/5 mL
- Promethazine hydrochloride suppository: 12.5 mg; 25 mg; 50 mg
- Promethazine hydrochloride oral syrup: 6.25 mg/5 mL
- Promethazine hydrochloride oral tablets: 12.5 mg, 25 mg, 50 mg

Promethazine should be given with food, water, or milk to decrease gastrointestinal discomfort when administered orally. The intramuscular injection must be into deep muscle tissue as the subcutaneous injection may damage tissue. Intravenous use should be performed with caution as severe tissue damage may also occur. Promethazine may be diluted before IV administration in selected patients to prevent promethazine-induced tissue necrosis.[13][14]

In pediatrics, promethazine has the same routes of administration as adults. However, dosing adjustments are necessary based on the weight of the patient and the indication. Utilizing the lowest dose of efficacy is recommended. There are no dosage adjustments for renal or hepatic impairment.

Allergy

The normal recommended oral dose is 25 mg administered before bed; however, alternatively, a patient can be given 6.25 to 12.5 mg may be taken three times before meals and on retiring, as necessary. After initiation of the treatment, the dose should be adjusted to the minimum dose to relieve the symptoms adequately in children or adults.

Motion Sickness

The average recommended adult dose is 25 mg, taken twice daily. The initial dose needs to be taken one-half to one hour before travel and repeated every eight to twelve hours later if needed. On following days of travel, 25 mg can be administered when up and repeated before the evening meal. Promethazine hydrochloride syrup, tablets, or rectal suppositories are administered at 12.5 to 25 mg twice daily for children.[7]

Nausea and Vomiting

Antiemetics are not recommended to treat vomiting of unknown etiology in children and adolescents. The average effective dose of promethazine for the active treatment of nausea and vomiting in children or adults is 25 mg. When the patients do not tolerate oral medication, it can be administered parenterally or rectally. Promethazine 12.5 mg to 25 mg may be repeated at four to six-hour intervals. For treatment of nausea and vomiting in children, the usual recommended dose is 0.5 mg/pound (bodyweight). The dose should be based on the patient's age, body weight, and the severity of nausea/vomiting. During surgery and the postoperative period, the recommended dose is 25 mg administered at four to six-hour intervals for prophylaxis of nausea and vomiting.[15]

Sedation

Promethazine relieves apprehension, inducing a quiet sleep from which the person can be easily aroused. Administer 12.5 mg to 25 mg promethazine orally or rectally at bedtime to provide sedation in children. Adults usually need 25 mg to 50 mg before surgery to relieve apprehension and produce quiet sleep for nighttime, obstetrical, or presurgical sedation. Children need 0.5 mg promethazine/pound (bodyweight) as a preoperative medication combined with adequately reduced narcotic/barbiturate and the required amount of an atropine-like drug. The usual recommended adult dosage is 50 mg promethazine with an adequately reduced dose of narcotic/barbiturate and the appropriate dose of a belladonna alkaloid. Postoperative sedation and adjunctive use with analgesic medicines may be achieved by the 12.5 mg to 25 mg promethazine in children and 25 mg to 50 mg promethazine in adults.

Adverse Effects

There are several potential adverse effects of promethazine administration related to its mechanism of action. The most common side effects include sedation, confusion, and disorientation, which may impair physical and mental abilities. However, in some cases, promethazine may paradoxically cause excitability, restlessness, or rare seizures.

Promethazine's anticholinergic properties may cause side effects, including blurred vision, xerostomia, dry nasal passages, dilated pupils, constipation, and urinary retention. Due to these effects, the American Geriatrics Society categorizes promethazine as a potentially inappropriate drug for the elderly.[16]

Promethazine's antidopaminergic properties may result in extrapyramidal symptoms, including pseudoparkinsonism, acute dystonia, akathisia, and tardive dyskinesia. Promethazine, therefore, may worsen symptoms in patients with Parkinson disease.[16]

Less common adverse effects that prescribers should be aware of include:

- The neuroleptic malignant syndrome may also have associations with promethazine usage, which manifests as increased body temperature, confusion or altered mental status, sweating, autonomic instability, and "lead pipe" rigidity.
- Cardiovascular side effects include arrhythmias and hypotension.
- Reports also exist of liver damage and cholestatic jaundice with its use.
- Bone marrow suppression, potentially resulting in agranulocytosis, thrombocytopenia, and leukopenia.
- Depression of the thermoregulatory mechanism resulting in hypothermia/hyperthermia.

Black Box Warnings

- In children less than two years old, reports exist of respiratory depression resulting in fatalities.[17] For this reason, there is a US black box warning on the use of promethazine in children less than two years of age.[18]
- Promethazine may cause serious tissue injury on injection, including gangrene, regardless of the route of administration. This effect may occur due to extravasation, unintentional intra-arterial administration, and intraneuronal or perineuronal infiltration, manifesting as burning, pain, erythema, edema, severe spasms, phlebitis, thrombophlebitis, venous thrombosis, sensory loss, paralysis, and palsies.[13][14]

Contraindications

Promethazine contraindications include patients with hypersensitivity to the drug, any drug components, or other phenothiazines. It is contraindicated in children under two years of age due to the risk of potentially fatal respiratory depression. Promethazine is contraindicated for subcutaneous or intra-arterial administration due to the risk of tissue damage. Contraindications also include comatose patients and patients with lower respiratory tract symptoms.[13][18]

Precautions/Warnings

Seizures: Promethazine should be used with caution in patients with seizure disorders or in persons taking concomitant medications, which may also lower seizure threshold (i.e., narcotics, local anesthetics, etc.)

Bone-Marrow Depression: Promethazine should be used cautiously in patients with bone marrow depression (Leukopenia and agranulocytosis). The risk is higher when it is coadministered with other known marrow-toxic agents.

Neuroleptic Malignant Syndrome: A potentially fatal NMS is reported in patients using promethazine alone or combined with other antipsychotic drugs. Clinical symptoms of NMS are hyperpyrexia, altered mental status, muscle rigidity, and autonomic instability (i.e., diaphoresis, irregular heartbeats or blood pressure, tachycardia, and cardiac dysrhythmias). The diagnostic evaluation of patients with NMS is complicated. It is important to identify patients when the clinical presentation includes untreated or inadequately treated EPS and serious medical conditions (e.g., pneumonia, systemic infection, etc.).[19]

Drug Interactions

Central-Nervous-System Depressants: Promethazine may intensify, increase, or prolong the sedative action of other CNS depressants, such as tricyclic antidepressants, sedatives/hypnotics, narcotics, narcotic analgesics, alcohol, general anesthetics, and tranquilizers; therefore, those medicines should be avoided or administered in a lower dosage to patients taking promethazine concomitantly. When promethazine is concomitantly administered, the dose of barbiturates should be reduced by at least 50%, and narcotics by 25% to 50%. Dosage must be individualized according to the patient's clinical presentation. High amounts of promethazine relative to narcotics may lead to motor hyperactivity and restlessness in the patient with pain, and these symptoms normally disappear with adequate pain management.

Anticholinergics: Concomitant use of other medicines with anticholinergic properties should be done with caution.

Monoamine Oxidase Inhibitors (MAOI): Concomitant use can result in drug interactions, including an increased incidence of EPS.

Drug/Laboratory Test Interactions: Some laboratory tests may be altered in patients receiving promethazine therapy. Diagnostic pregnancy test results based on immunological reactions of HCG and anti-HCG could result in false-positive or false-negative interpretations. Also, a glucose tolerance test may report falsely elevated blood glucose levels in patients receiving promethazine.

Monitoring

Due to promethazine's side effect profile, any medical personnel who administers the drug should be aware of potential side effects. Healthcare providers need to monitor patients for burning or pain at the injection site, phlebitis, blistering, or swelling, which may indicate tissue damage.[14] Extrapyramidal and anticholinergic side effects also require monitoring, and providers should be notified immediately if they appear. Promethazine is also a CNS depressant, so safety measures such as side rail up and call light within reach should be in place.

Toxicity

The main feature of promethazine toxicity is CNS depression, tachycardia, respiratory depression, and delirium. For most overdoses, supportive care and monitoring are the treatment as there is no known antidote. Significant overdoses with profoundly depressed mental status or coma may require airway support, hemodynamic monitoring, and a higher level of care. Some studies have shown that early administration of charcoal within 2 hours may be beneficial. However, the evidence is still lacking, and further studies are necessary.[11] Because promethazine can reverse epinephrine's vasopressor effect, epinephrine should not be used for hypotension associated with promethazine overdose.

Management of NMS

Immediate discontinuation of promethazine, antipsychotic drugs, and other drugs is not essential to concurrent treatment. Intensive symptomatic treatment and medical monitoring Provide treatment for concomitant serious medical problems, where specific treatments are available. There are no general recommendations or specific pharmacological treatment protocols for uncomplicated NMS. However, as reoccurrences of NMS have been reported with phenothiazines class medicines, the reintroduction of promethazine should be carefully done.

Enhancing Healthcare Team Outcomes

Promethazine is a relatively safe medication that is very useful in several circumstances, such as nausea/vomiting, allergic conditions, prevention of motion sickness, and pre/post-operative or obstetric sedation. However, any prescriber of the medication should be aware of the potential side effects and contraindications. In addition, providers must be mindful of the two black box warning associated, which include the potential to cause severe tissue damage with IM/IV injection and potentially fatal respiratory depression in children under two years of age.

Therefore, close interprofessional coordination between providers (MDs, DOs, NPs, PAs), nurses, pharmacists, and other healthcare workers is necessary to improve patient outcomes and decrease adverse events when using promethazine. Pharmacists can reconcile the patient's medication profile and report to the nurse or prescribing clinician if any interactions exist. Nurses will often be the first to see the patient and monitor for apparent adverse effects, either reporting to the prescriber or checking with the pharmacist regarding the adverse event profile of promethazine. Nursing is also in charge of IV administration to have first-line exposure to issues such as extravasation and can inform the clinician. The entire interprofessional healthcare team must collaborate, each discipline bringing its expertise to support positive outcomes. [Level 5]